

Project Design Phase

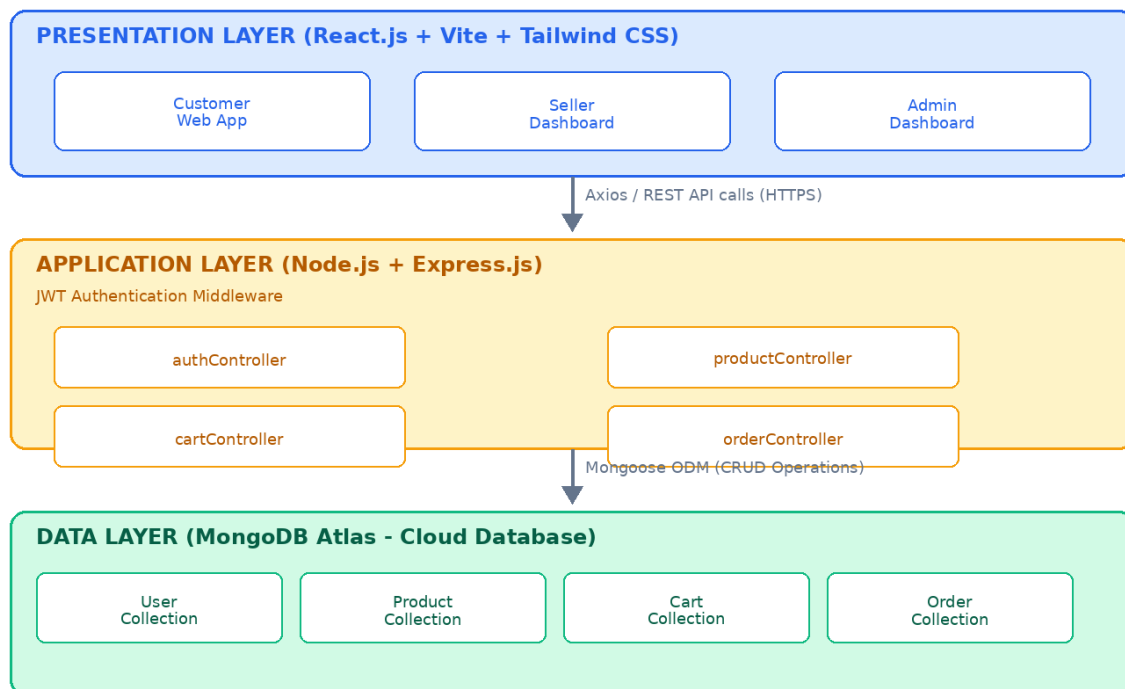
Solution Architecture

Date	12 July 2026
Team ID	Buyya Vaishnavi
Project Name	ShopEZ
Maximum Marks	4 Marks

Solution Architecture:

The ShopEZ application utilizes a 3-layer MERN architecture which comprises of React, Express, and MongoDB. Three types of users can access the platform, namely Customers, Sellers, and Admins. Customers browse products and place orders, Sellers manage their own product listings and view orders for their products, and Admins oversee all users, products and orders across the system. JWT is employed in the authentication process of accessing the application, with data stored in MongoDB.

Solution Architecture Diagram:



The proposed ShopEZ solution architecture uses a three-tier MERN stack along with MVC (Model View Controller) pattern for designing a clearly separated front-end, back-end, and database layers. The front-end is made with React 19 and TypeScript, powered by Vite, and is an implementation of the View layer where users (Customers, Sellers, Administrators) can access and interact with the application using its interfaces.

Node.js and Express.js REST API server serves as the Controller and handles all the requests received from the client side using Axios HTTP requests. For security reasons, secured routes have been protected using JWT Authentication Middleware. Depending on the request, it routes the request further to the required controller module including authController, productController, cartController, and orderController.

The controllers work with the database using Mongoose ODM to carry out CRUD operations within collections like User, Product, Cart, and Order. These collections constitute the Model and determine the structure and relations within the application data. The database handles the requested data and sends back responses to the backend that are rendered by the frontend interface. Using the MVC-based MERN architecture, the ShopEZ application can be made highly secure and scalable while being able to manage an online marketplace for customers, sellers, and admins effectively.